Huang Fang

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Academic Appointments and Education

2026-Present	**Assistant Professor** State Key Laboratory of Surface Physics, Fudan University
2022-2026	**Postdoctoral Researcher**, Physics, Fudan University Advisor: Prof. Peng Tan
2016-2022	**Ph.D.**, Physics, Brandeis University Advisors: Prof. W. Benjamin Rogers; Prof. Michael F. Hagan
2012-2016	**B.S.**, Optical Information Science and Technology, University of Science and Technology of China

Research Interests

Interfacial physics and low-dimensional assemblies, including geometrically frustrated and confined soft matter and nonequilibrium complex systems. Enhanced-sampling methods for molecular simulations and AI-assisted inverse design and optimization of programmable intelligent materials.

Publications

† These authors contributed equally

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1. **Huang Fang**†, Xiaotian Li†, Wensi Sun†, Chengxin Wang, Nuo Chen, Yining Gan, Jiping Huang, Yuqiang Ma*, Hajime Tanaka*, Peng Tan*, "Dual-symmetry-guided assembly of complex lattices", *Nature* **652**, 635–642 (2026).
2. Qiong Gao†, **Huang Fang**†, Dong Xiang, Yanshuang Chen, Hajime Tanaka* and Peng Tan*, "Impact of impurities on crystal growth", *Nat. Phys.* **21**, 938–946 (2025).
3. **Huang Fang**†, Qiong Gao†, Yujie Rong†, Yanshuang Chen, Jiping Huang, Hua Tong, Zhihong Nie*, Hajime Tanaka*, Wei Li*, and Peng Tan*, "Dynamic and asymmetric colloidal molecules", *Nat. Commun.* **16**, 2819 (2025).
4. **Huang Fang**, Michael F. Hagan, W. Benjamin Rogers*, "Two-step crystallization and solid–solid transitions in binary colloidal mixtures", *PNAS* **117**, 27927-27933 (2020).
5. Xiaoxia Li†, **Huang Fang**†, Krongtum Sankaewtong†, Minhuan Li, Yanshuang Chen, Jiping Huang, Ran Ni*, Hajime Tanaka*, and Peng Tan*, "Phase Reentrances and Solid Deformations in Confined Colloidal Crystals", *Phys. Rev. Lett*. **132**, 018202 (2024).
6. Quanzheng Deng†, **Huang Fang**†, Xueliang Zhang, Peng Tan*, Lu Han*, "Mesostructured Dodecagonal Quasicrystal: the Interplay Between Temperature Regulation and Kinetic Structural Transition", *Small* **21**, 2502064 (2025).
7. **Huang Fang**, Botond Tyukodi, W. Benjamin Rogers, Michael F. Hagan*, "Polymorphic self-assembly of helical tubules is kinetically controlled", *Soft Matter* **18**, 6716-6728 (2022).
8. **Huang Fang**, Peng Tan, "Packing Problems in Soft Matter Physics: Fundamentals and Applications, Chapter 17: Packing of Charged Spheres", *Royal Society of Chemistry* 433-450 (2025).
9. Pepijn G. Moerman, **Huang Fang**, Thomas E. Videbæk, W. Benjamin Rogers*, and Rebecca Schulman*, "A simple method to reprogram the binding specificity of DNA-coated colloids that crystallize", *Soft Matter* **19,** 8779-8789 (2023).

10. Thomas E. Videbaek, **Huang Fang**, Daichi Hayakawa, Botond Tyukodi, Michael F. Hagan, W. Benjamin Rogers*, "Tiling a tubule: How increasing complexity improves the yield of self-limited assembly", *Journal of Physics: Condensed Matter* **34**, 134003 (2022).
11. Simon Merminod, John R. Edison, **Huang Fang**, Michael F. Hagan, W. Benjamin Rogers*, "Avidity and surface mobility in multivalent ligand–receptor binding", *Nanoscale* **13**, 12602-12612 (2021).
12. Yifan Gao, Shenxin Xia, **Huang Fang**, Zhebin Zhang, Dong Yang, Tongtao Li*, Wei Li*, Angang Dong*, "General Synthesis of 2D Mesoporous Multimetal Spinel Supracrystals for High-Efficiency Electrochemical Applications", *Angew. Chem. Int. Ed.* **64**, e202510954 (2025).
13. Daichi Hayakawa, Thomas E. Videbæk, Douglas M. Hall, **Huang Fang**, Christian Sigl, Elija Feigl, Hendrik Dietz, Seth Fraden, Michael F. Hagan, Gregory M. Grason, W. Benjamin Rogers*, "Geometrically programmed self-limited assembly of tubules using DNA origami colloids", *PNAS* **119**: e2207902119 (2022).
14. Wei Zhang, Wenxiong Shi, Jinxiu Wang, Kerun Zhu, Yong Tian, Haili He, Linlin Duan, **Huang Fang**, Yupu Liu, Biwei Wang, Yuzhu Ma, Tiancong Zhao, Xingmiao Zhang, Zirui Lv, Lianhai Zu, Yun Tang, Angang Dong, Biao Kong*, Dongyuan Zhao*, "A Library of Nanocrystal-Inserted Ordered Mesoporous Frameworks with Ultrahigh Density and Spatial Dispersity", *Adv. Funct. Mater.* **34**, 2401814 (2024).

In preparation

1. **Huang Fang**, Peng Tan*, "Curvature-dependent mesoscale morphologies in two-dimensional crystals".
2. Yuyue Gong, **Huang Fang***, Peng Tan*, "Competing frustrations drive morphology selection and pathway bifurcation in colloidal supraspheres".
3. Hao Wang†, **Huang Fang**†, Chuncheng Li, Yining Gan, Siyu Wan, Yutong Gao, Jingxing Jiao, Xinran Xu, Siyuan Liu, Mingxuan Bian, Longfei Lv, Yifan Gao, Dong Yang, Tongtao Li, Zhaochuan Fan, Peng Tan*, Angang Dong*, "Octagonal quasicrystalline nanoparticle superlattices enabled by asymmetric attraction".

Research Funding

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2025 China Postdoctoral Science Foundation, General Financial Grant.
Grant No. 2025M773335; RMB 80,000.

2022 China Postdoctoral Science Foundation, General Financial Grant (Second Class).
Grant No. 2022M720811; RMB 80,000.

Talks and Posters

1. “介尺度阻挫下的曲面二维晶体”, 复旦大学物理学系 2025 年度学术年会，上海，2025 年 4 月 24 日
2. “Two-dimensional crystal on a sinusoidally curved manifold”, *APS March Meeting Hong Kong Satellite Session, Hong Kong, March 18, 2025*
3. “动态“类分子”胶体团簇的非对称性”, 第十三届全国软物质与生命物质物理学术会议，西安，2024 年 3 月 31 日
4. “Understanding the mechanism of self-limited assembly of tubules using Monte Carlo simulations”, *APS March Meeting, Chicago, IL, March 15, 2022*
5. “Exploring nucleation pathways and solid-solid phase transitions in two-dimensional binary colloidal crystallization”, *APS March Meeting, Online, August 18, 2021*
6. “Exploring nucleation pathways and solid-solid phase transitions in two-dimensional binary colloidal crystallization”, *Gordon Research Conference poster session, New London, NH, August 13-14, 2019*
7. “Exploring nucleation pathways and solid-solid phase transitions in two-dimensional binary colloidal crystallization”, *Gordon Research Seminar, New London, NH, August 11, 2019*
8. “Exploring nucleation pathways and solid-solid phase transitions in two-dimensional binary colloidal crystallization”, *APS March Meeting, Boston, MA, March 4, 2019*
9. “Exploring nucleation pathways and solid-solid phase transitions in two-dimensional binary colloidal crystallization”, *APS March Meeting, Los Angeles, CA, March 7, 2018*

Teaching Experience

2017-2018 **Teaching Assistant**, Electronic Laboratory (PHYS 29), Brandeis University

2016-2017 **Teaching Assistant**, Introductory Laboratory (PHYS 18), Brandeis University